

Phase 2: Pre-processing and Model Training

A Comparative Analysis of Machine Learning Architectures for Satellite-Based Surface Monitoring in Hunga Tonga–Hunga Ha’apai

Daniel Fernando Gonzalez Cortés
Andrés David Pérez Cely
María de los Angeles de la Rosa Ortiz
Isaac Alejandro Jánica Piñeros

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Abstract

This second phase documents the technical preparation of the supervised classification workflow for the Hunga Tonga–Hunga Ha’apai case study. The work focuses on the preprocessing of Sentinel-2 multispectral bands, the construction of a tabular training dataset, and the initial training and evaluation of five machine learning models: Decision Tree, Support Vector Machine, Artificial Neural Network, K-Nearest Neighbor, and Naive Bayes. Ten Sentinel-2 spectral bands were used as predictor variables, while five land-cover and atmospheric categories were defined as target classes: ocean water, bare soil, vegetation, cloud, and cloud shadow. The extracted dataset contained 131,840 pixel-level samples and was later balanced to 2,456 samples per class for model training. The initial results show that K-Nearest Neighbor achieved the highest performance, with an overall accuracy of 0.9967, a Kappa coefficient of 0.9959, and a macro F1-score of 0.9967.

1 Introduction

This document corresponds to the second phase of the final project, focused on preprocessing and model training. The first phase defined the study area, selected the Sentinel-2 image, identified the classification classes, and created the training polygons. This phase continues the workflow by transforming the multispectral raster information into a tabular dataset suitable for supervised machine learning.

The selected case study is the Hunga Tonga–Hunga Ha’apai volcanic system. The image used corresponds to a Sentinel-2 scene acquired on November 18, 2021. This image was selected because the island is still visible before the major January 2022 eruption and because the scene contains enough spectral diversity to classify ocean water, exposed land, vegetation, clouds, and cloud shadows.

The main objective of this phase is to document the creation of the training dataset and evaluate the initial performance of five machine learning architectures using standard classification metrics.

2 Pre-processing Workflow

The preprocessing workflow was implemented mainly in Python. QGIS was used to support the visual interpretation of the image and the digitization of the training polygons. The remaining technical operations, including band reading, resampling, pixel extraction, tabular dataset construction, model training, hyperparameter optimization, and metric computation, were performed programmatically.

The complete preprocessing workflow included the following steps:

1. Load the selected Sentinel-2 bands.
2. Use the 10 m bands as the spatial reference grid.
3. Resample the 20 m bands to match the 10 m grid.

4. Load the training polygons created in QGIS.
5. Extract the spectral values of the pixels located inside each polygon.
6. Convert pixel coordinates into latitude and longitude.
7. Assign the corresponding class label to each extracted pixel.
8. Export the resulting dataset as a TSV file.

3 Sentinel-2 Bands Used

The classification dataset was built using ten Sentinel-2 spectral bands. Four bands were originally available at 10 m spatial resolution, while six bands were originally available at 20 m and were resampled to the 10 m reference grid.

Table 1: Sentinel-2 bands used for model training.

Band	Description	Original resolution
B02	Blue	10 m
B03	Green	10 m
B04	Red	10 m
B05	Red Edge 1	20 m
B06	Red Edge 2	20 m
B07	Red Edge 3	20 m
B08	Near Infrared	10 m
B8A	Narrow Near Infrared	20 m
B11	Short-Wave Infrared 1	20 m
B12	Short-Wave Infrared 2	20 m

The 20 m bands were resampled using the spatial grid of the 10 m bands. This step was necessary to ensure that all spectral values corresponded to the same pixel locations during feature extraction.

4 Training Classes

Five classes were defined based on the visible surface and atmospheric conditions present in the selected image. These classes were used as the categorical labels for supervised learning.

Table 2: Classification classes.

Class ID	Class name	Description
1	Ocean_water	Open ocean water surrounding the island.
2	Bare_soil	Exposed land, volcanic soil, sand, or non-vegetated surface.
3	Vegetation	Vegetated areas identified using natural and false color interpretation.
4	Cloud	Bright cloud features visible in the optical image.
5	Cloud_shadow	Dark shadow areas associated with cloud cover.

5 Training Polygons

The training polygons were created in QGIS over visually homogeneous areas. The purpose of these polygons was to provide ground-truth samples for the supervised classification process. The polygons were digitized while avoiding mixed areas such as coastlines, cloud borders, transitions between vegetation and soil, and ambiguous shadow zones.

Table 3: Number of training polygons by class.

Class name	Number of polygons
Ocean_water	17
Cloud	14
Bare_soil	14
Vegetation	14
Cloud_shadow	14

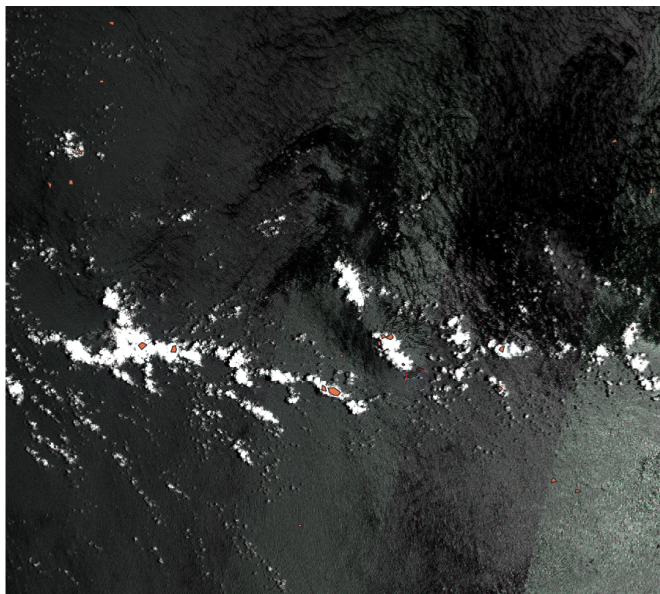


Figure 1: Training polygons digitized over the Sentinel-2 image.

6 TSV Training Dataset

The training polygons were used to extract pixel-level spectral signatures from the Sentinel-2 bands. Each row of the resulting dataset represents one pixel located inside a training polygon.

The exported file was named:

`training_data.tsv`

The dataset includes geographic coordinates, spectral features, and class labels.

Table 4: Structure of the exported TSV dataset.

Column	Description
Latitude	Latitude of the pixel center.
Longitude	Longitude of the pixel center.
B02	Blue band value.
B03	Green band value.
B04	Red band value.
B05	Red Edge 1 band value.
B06	Red Edge 2 band value.
B07	Red Edge 3 band value.
B08	Near Infrared band value.
B8A	Narrow Near Infrared band value.
B11	Short-Wave Infrared 1 band value.
B12	Short-Wave Infrared 2 band value.
class_id	Numerical class identifier.
class_name	Categorical class label.

7 Original Dataset Distribution

The extraction process produced a total of 131,840 pixel samples. However, the distribution was not balanced because clouds and ocean water occupied larger areas within the training polygons than vegetation and bare soil.

Table 5: Original distribution of extracted pixel samples.

Class name	Extracted pixels
Cloud	63,930
Ocean_water	50,572
Cloud_shadow	11,861
Bare_soil	3,021
Vegetation	2,456
Total	131,840

This imbalance could bias the models toward the majority classes. For this reason, a balanced version of the dataset was created before model training.

8 Dataset Balancing

The dataset was balanced by randomly sampling the same number of pixels from each class. Since Vegetation was the minority class with 2,456 samples, this value was used as the reference number for all classes.

Table 6: Balanced dataset used for training and testing.

Class name	Samples used
Bare_soil	2,456
Cloud	2,456
Cloud_shadow	2,456
Ocean_water	2,456
Vegetation	2,456
Total	12,280

The balanced dataset was then divided into training and testing subsets using a stratified split.

Table 7: Training and testing split.

Subset	Number of samples
Training set	9,210
Testing set	3,070

9 Machine Learning Models

Five supervised machine learning models were implemented and compared:

1. **Decision Tree:** a rule-based classifier that separates classes using hierarchical decision rules.
2. **Support Vector Machine:** a margin-based model that separates classes using hyperplanes in the feature space.
3. **Artificial Neural Network:** a multilayer model capable of learning nonlinear relationships between spectral bands and classes.
4. **K-Nearest Neighbor:** a non-parametric classifier that assigns each sample to the most common class among its nearest neighbors.
5. **Naive Bayes:** a probabilistic classifier based on Bayes' theorem and an independence assumption among features.

10 Hyperparameter Optimization

A hyperparameter optimization process was performed using grid search and cross-validation. The objective was to select the best configuration for each model based on macro F1-score.

Table 8: Best hyperparameters obtained for each model.

Model	Best hyperparameters
Decision Tree	criterion = entropy; max_depth = 20; min_samples_split = 5
SVM	C = 10; gamma = scale; kernel = linear
ANN	activation = tanh; hidden_layer_sizes = (100,); learning_rate_init = 0.01
KNN	metric = euclidean; n_neighbors = 3; weights = distance
Naive Bayes	var_smoothing = 1e-06

11 Evaluation Metrics

The models were evaluated using the following metrics:

- **Overall Accuracy:** proportion of correctly classified samples.
- **Kappa Coefficient:** agreement between predictions and reference labels beyond chance.
- **Precision:** proportion of predicted samples that were correct.
- **Recall:** proportion of actual samples correctly identified.
- **F1-score:** harmonic mean between precision and recall.

Macro-averaged metrics were used to give equal importance to each class.

12 Model Comparison Results

The results show that KNN achieved the highest performance among the five evaluated architectures, followed by Decision Tree, SVM, ANN, and Naive Bayes.

Table 9: Model comparison using the balanced test set.

Model	Accuracy	Kappa	Precision	Recall	F1-score
KNN	0.9967	0.9959	0.9968	0.9967	0.9967
Decision Tree	0.9941	0.9927	0.9941	0.9941	0.9941
SVM	0.9853	0.9817	0.9862	0.9853	0.9853
ANN	0.9847	0.9809	0.9855	0.9847	0.9847
Naive Bayes	0.8814	0.8518	0.8785	0.8814	0.8794

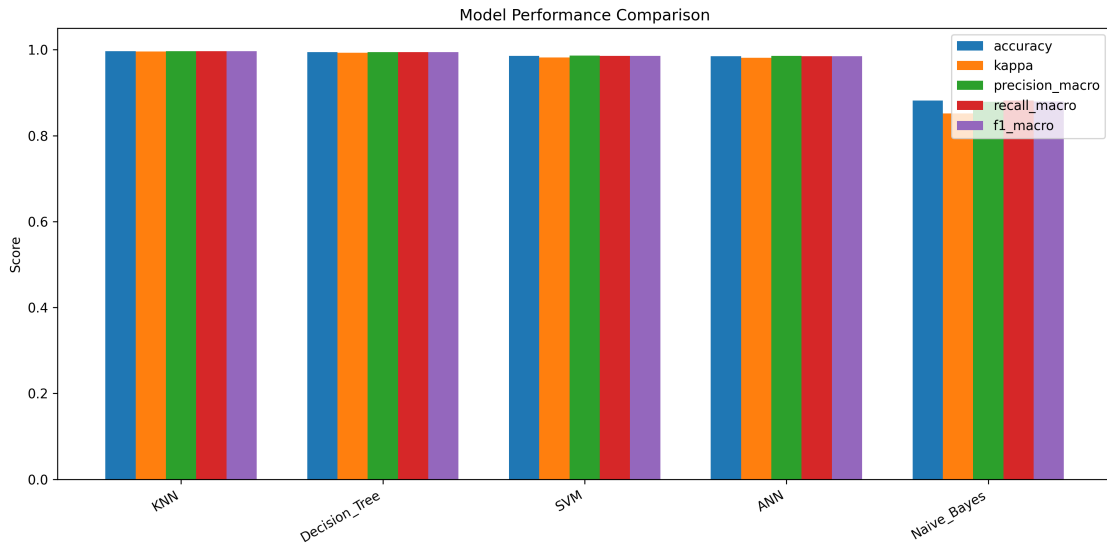


Figure 2: Comparison of model performance metrics.

13 Confusion Matrix of the Best Model

Since KNN achieved the best performance, its confusion matrix was analyzed in greater detail. The results show that almost all samples were correctly classified. Most of the classification errors occurred between Ocean_water and Cloud_shadow, which is reasonable because both classes can present low reflectance values in visible bands.

Table 10: Confusion matrix for the KNN model.

Actual / Predicted	Bare_soil	Cloud	Cloud_shadow	Ocean_water	Vegetation
Bare_soil	611	0	0	3	0
Cloud	0	614	0	0	0
Cloud_shadow	0	0	613	1	0
Ocean_water	0	0	6	608	0
Vegetation	0	0	0	0	614

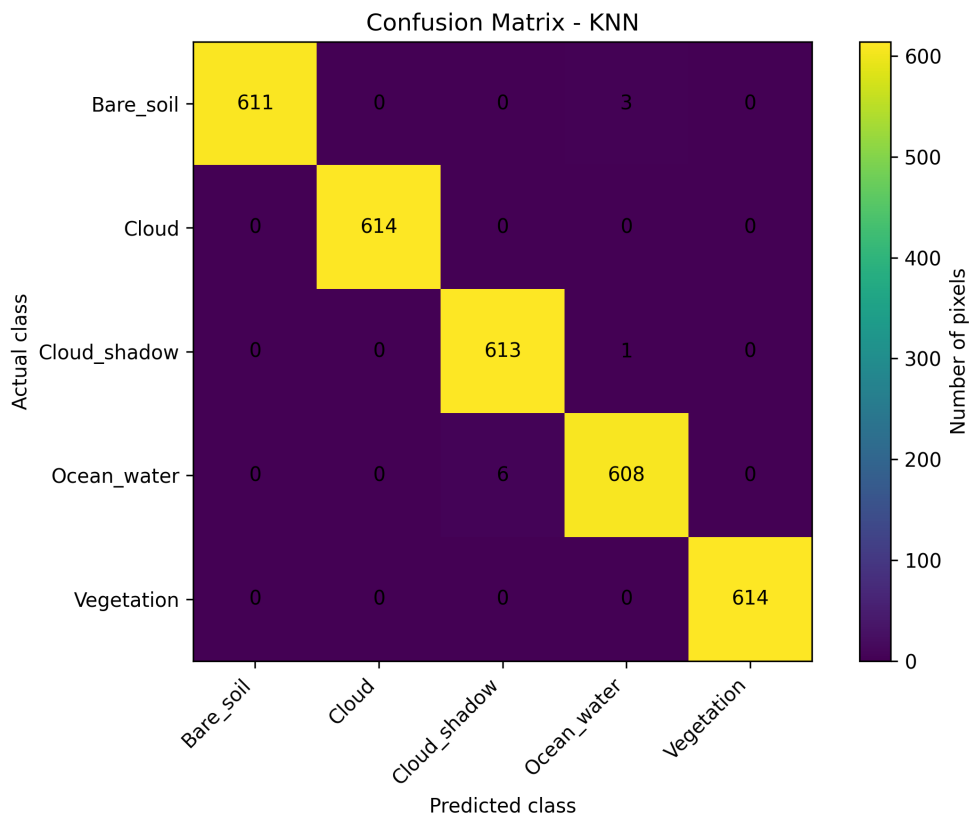


Figure 3: Confusion matrix for the KNN classifier.

14 Initial Interpretation

The initial model evaluation indicates that the spectral bands selected for the project provide enough information to distinguish the five classes. The best performance was obtained by KNN, with an overall accuracy of 0.9967, a Kappa coefficient of 0.9959, and a macro F1-score of 0.9967.

Decision Tree also achieved high performance, suggesting that several classes can be separated using clear spectral thresholds. SVM and ANN performed similarly and produced strong results, although slightly below KNN and Decision Tree. Naive Bayes obtained the lowest performance, which may be explained by its independence assumption, since Sentinel-2 spectral bands are often highly correlated.

The main source of confusion occurred between Ocean_water and Cloud_shadow. This result is expected because both classes can appear dark in the visible bands and may have similar spectral behavior in certain parts of the image.

15 Area Summary

Although full-scene raster prediction belongs mainly to the next phase, an initial area summary was generated after applying the classification workflow to the complete Sentinel-2 scene. This information is included only as a preliminary reference.

Table 11: Preliminary area by class over the processed Sentinel-2 scene.

Class name	Pixel count	Area ha	Area km ²
Ocean_water	113,631,591	1,136,315.91	11,363.1591
Cloud_shadow	4,142,218	41,422.18	414.2218
Cloud	1,951,958	19,519.58	195.1958
Bare_soil	827,967	8,279.67	82.7967
Vegetation	6,661	66.61	0.6661

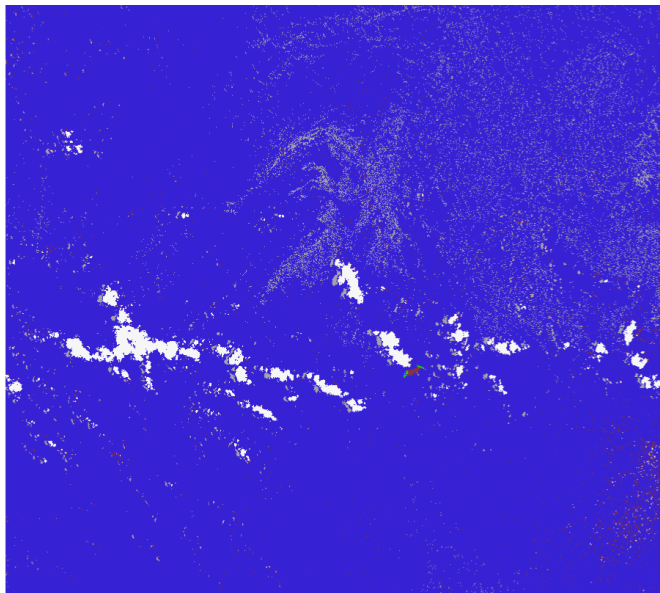


Figure 4: Preliminary classified area by class over the full Sentinel-2 scene.

16 Conclusion

This second phase successfully completed the preprocessing and training stage of the supervised classification workflow. The Sentinel-2 bands were integrated into a common 10 m spatial grid, and pixel-level spectral values

were extracted from the training polygons to create the `training_data.tsv` file. The original dataset contained 131,840 extracted samples, which were balanced to 2,456 samples per class before model training.

Five machine learning architectures were implemented and optimized using grid search. The best-performing model was KNN, which achieved an overall accuracy of 0.9967, a Kappa coefficient of 0.9959, and a macro F1-score of 0.9967. These results indicate that the selected spectral bands and training samples are suitable for distinguishing ocean water, bare soil, vegetation, cloud, and cloud shadow classes. The next phase will focus on full raster inference, thematic GeoTIFF generation, map visualization, and final area quantification.